

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K} = (O, \mathcal{B})$ where $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

1. (1 point) Select all that apply. Which of the following equalities hold true for the oriented Euclidean plane?

A. $\cos \angle(\mathbf{a}, \mathbf{b}) = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{|\mathbf{a}| \cdot |\mathbf{b}|}$ B. $\sin \angle_{\text{or}}(\mathbf{a}, \mathbf{b}) = \frac{[\mathbf{a}, \mathbf{b}]}{|\mathbf{a}| \cdot |\mathbf{b}|}$ C. $\tan \angle_{\text{or}}(\mathbf{a}, \mathbf{b}) = \frac{[\mathbf{a}, \mathbf{b}]}{\langle \mathbf{a}, \mathbf{b} \rangle}$ D. $\cot \angle_{\text{or}}(\mathbf{a}, \mathbf{b}) = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{[\mathbf{a}, \mathbf{b}]}$

2. (1 point) The distance between the opposite sides parallel to $\overrightarrow{AB}$ of a parallelogram spanned by the vectors $\overrightarrow{AB}(6, 0, 1)$ and $\overrightarrow{AC}(1.5, 2, 1)$ is _____.

Setup 1: Consider the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \quad [D]_{\mathcal{K}} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

3. (1 point) Consider Setup 1 above. The area of the triangle ABC is $\text{Area}(ABC) = \underline{\hspace{2cm}}$.
4. (1 point) Consider Setup 1 above. The volume of the tetrahedron $ABCD$ is $\text{Vol}_{\text{tetra}}(ABCD) = \underline{\hspace{2cm}}$.
5. (1 point) Consider Setup 1 above. A direction vector for the common perpendicular line of the lines AD and BC is _____.
6. (1 point) Consider Setup 1 above. The distance between the lines AB and CD is _____.
7. (1 point) Consider Setup 1 above. Cartesian equations for the common perpendicular line of the lines AB and CD are $\left\{ \begin{array}{l} \underline{\hspace{2cm}} \\ \underline{\hspace{2cm}} \end{array} \right.$.
8. (1 point) Consider Setup 1 above. The distance from the point D to the plane containing the triangle ABC is _____.
9. (1 point) Select all that apply. The vector product is
A. bilinear B. skew-symmetric C. positive definite D. symmetric
10. (1 point) Consider the points $A(2, 1, -1)$, $B(3, 0, 1)$, $C(2, -1, 3)$ and $D \in Oy$. The coordinates of a point D , for which the volume of the tetrahedron $ABCD$ is 5, are _____.

Answers:

1. A, B, C, D
2. $\frac{\sqrt{673}}{2\sqrt{37}}$
3. $\sqrt{5}$
4. 1
5. $(5, -3, 1)$
6. 1
7. $\begin{cases} 3y - 4 = 0 \\ z - 1 = 0 \end{cases}$
8. $\frac{3}{\sqrt{5}}$
9. A, B
10. $(0, -7, 0)$ or $(0, 8, 0)$